

# sheLLaMa as a Cloud AI Cost Estimation Platform

Measuring Local LLM Usage to Project Cloud Provider Spend

sheLLaMa Project

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## Abstract

Organizations adopting AI face a critical question: what would our workload cost on cloud AI providers? sheLLaMa answers this by running inference locally on commodity hardware while simultaneously tracking every token consumed and projecting costs across 28+ cloud models from OpenRouter, Amazon Bedrock, OpenAI, Anthropic, Google, xAI, and Meta. This paper describes the architecture, methodology, and practical use of sheLLaMa as a cloud AI cost estimation and capacity planning tool.

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## 1 Introduction

Cloud AI services charge per token—typically split into input (prompt) and output (completion) tokens. Pricing varies by orders of magnitude: from \$0.035/M tokens (Amazon Nova Micro) to \$75/M tokens (Claude Opus 4 output). For organizations evaluating AI adoption, the question is not whether AI is useful, but whether the cost is justified at their usage volume.

sheLLaMa provides a zero-cost way to answer this question:

1. Run your actual AI workloads locally (code generation, chat, analysis, shell automation)
2. The platform counts every prompt and response token consumed
3. Real-time cost projections show what the same usage would cost on each cloud provider
4. Benchmarking compares local model quality against cloud alternatives

The result is a data-driven cost estimate based on *actual usage patterns*, not hypothetical projections.

## 2 The Cost Estimation Problem

### 2.1 Why Estimation is Hard

Cloud AI pricing is complex:

- **Asymmetric pricing:** Input and output tokens have different rates (e.g., Claude 4 Sonnet: \$3/M input, \$15/M output)
- **Variable output length:** The same prompt can produce vastly different response lengths depending on the model and task
- **Workload mix:** Organizations use AI for diverse tasks—chat, code generation, document analysis, shell automation—each with different token profiles
- **Rapid price changes:** Cloud providers update pricing frequently; static spreadsheet estimates become stale
- **Hidden costs:** Rate limiting, retries, conversation context accumulation, and cache misses all inflate real-world spend

### 2.2 sheLLaMa's Approach

Rather than estimating from assumptions, sheLLaMa measures actual usage:

- Every request records `prompt_tokens` and `response_tokens` from Ollama's response meta-data
- Token counts are accumulated per client IP, per task type, and per API key
- Cost projections use **live pricing** fetched from OpenRouter's API and AWS Pricing API
- Static fallback prices are maintained for when APIs are unreachable
- Benchmark tokens are excluded from cost projections to reflect real usage only

### 3 Architecture for Cost Tracking

#### 3.1 Token Recording Pipeline

Every request flowing through the frontend load balancer is instrumented:

Listing 1: Token recording in `proxy_request`

```
record_ip_tokens(
    ip=client_ip,
    tokens=result.get('total_tokens', 0),
    task_type=task_type,
    prompt_tokens=result.get('prompt_tokens', 0),
    response_tokens=result.get('response_tokens', 0),
    cloud_fallback=result.get('cloud_fallback', False),
    key_name=get_key_name()
)
```

Tokens are stored in three dimensions:

- **Cumulative totals** (persisted to disk, survive restarts)
- **Per-IP history** (timestamped entries for time-range queries)
- **Per-task breakdown** (chat, generate, explain, analyze, image)

#### 3.2 Pricing Data Sources

| Source          | Models Covered                                               | Refresh           |
|-----------------|--------------------------------------------------------------|-------------------|
| OpenRouter API  | Claude, GPT-4o, Gemini, Grok, Llama, Nova (15 models)        | Per /test request |
| AWS Pricing API | Bedrock on-demand: Nova, Llama, DeepSeek, Mistral (9 models) | Per /test request |
| Static fallback | All 28 models (hardcoded)                                    | Code updates      |

Table 1: Pricing data sources and coverage

The system attempts live pricing first. If OpenRouter or AWS is unreachable (offline environments), static fallback prices are used. The response always includes `pricing_source` (`openrouter` or `static`) so users know the data freshness.

#### 3.3 Cost Calculation

For each provider, the cost formula is:

$$\text{Cost} = \frac{\text{prompt\_tokens} \times P_{\text{input}}}{1,000,000} + \frac{\text{response\_tokens} \times P_{\text{output}}}{1,000,000}$$

where  $P_{\text{input}}$  and  $P_{\text{output}}$  are per-million-token rates.

## 4 Supported Cloud Providers and Models

### 4.1 OpenRouter Models (via API)

| Model             | Input (\$/M) | Output (\$/M) |
|-------------------|--------------|---------------|
| Claude 4 Sonnet   | 3.00         | 15.00         |
| Claude 4 Haiku    | 1.00         | 5.00          |
| Claude 3.5 Sonnet | 3.00         | 15.00         |
| GPT-4o            | 2.50         | 10.00         |
| GPT-4o mini       | 0.15         | 0.60          |
| OpenAI o3         | 2.00         | 8.00          |
| OpenAI o4-mini    | 1.10         | 4.40          |
| Gemini 2.5 Pro    | 1.25         | 10.00         |
| Gemini 2.5 Flash  | 0.30         | 2.50          |
| Grok 3            | 3.00         | 15.00         |
| Grok 3 mini       | 0.30         | 0.50          |
| Llama 3.1 70B     | 0.40         | 0.40          |
| Amazon Nova Pro   | 0.80         | 3.20          |
| Amazon Nova Lite  | 0.06         | 0.24          |
| Amazon Nova Micro | 0.04         | 0.14          |

Table 2: OpenRouter model pricing (static fallback values; live prices may differ)

### 4.2 Amazon Bedrock Models (via AWS Pricing API)

| Model             | Input (\$/M) | Output (\$/M) |
|-------------------|--------------|---------------|
| Claude Opus 4     | 15.00        | 75.00         |
| Claude 4 Sonnet   | 3.00         | 15.00         |
| Claude 3.5 Sonnet | 3.00         | 15.00         |
| Claude 3.5 Haiku  | 0.80         | 4.00          |
| Nova Pro          | 0.80         | 3.20          |
| Nova Lite         | 0.06         | 0.24          |
| Nova Micro        | 0.035        | 0.14          |
| Nova Premier      | 2.50         | 12.50         |
| Llama 4 Maverick  | 0.24         | 0.97          |
| Llama 4 Scout     | 0.17         | 0.66          |
| Llama 3.3 70B     | 0.72         | 0.72          |
| DeepSeek R1       | 1.35         | 5.40          |
| Mistral Large 3   | 0.50         | 1.50          |

Table 3: Amazon Bedrock on-demand pricing (us-east-1, static fallback values)

## 5 Cost Estimation Interfaces

### 5.1 Admin Console — Costs Page

The web-based costs page (`/costs`) provides:

- **Time range filtering:** Day, week, month, year, all-time, or custom date range
- **Hypothetical costs:** What your total local usage would cost on each cloud provider

- **Actual fallback spend:** Real costs incurred when cloud fallback was triggered
- **Cached savings:** Tokens served from cache that would have cost money on cloud
- **Visual indicators:** Cheapest provider highlighted green, most expensive in red
- **Bedrock section:** Separated with AWS orange styling for clarity
- **Live pricing source:** Shows whether prices are from OpenRouter or static fallback

## 5.2 CLI Benchmarking

The `,test` command provides per-request cost estimates:

Listing 2: CLI benchmark with cost estimates

```
$ ,test all
Testing 4 models with default prompt...

Model           Time    Prompt  Response  Total   tok/s
qwen2.5-coder:3b 12.3s   142     387       529    43.1
qwen2.5-coder:7b 34.7s   142     412       554    16.0
deepseek-coder:6.7b 38.2s  142     395       537    14.1
qwen2.5-coder:14b 97.1s   142     445       587     6.0

Cloud cost estimates (avg 142 prompt + 410 response tokens):
Provider          Input          Output          Total
Claude 4 Sonnet   $0.000426     $0.006150     $0.006576
GPT-4o            $0.000355     $0.004100     $0.004455
Amazon Nova Micro $0.000005     $0.000057     $0.000062
...
```

## 5.3 REST API

Listing 3: Cost estimation via API

```
# Get cumulative cost projections
curl http://server:5000/cloud-costs

# Get costs for a specific time range
curl "http://server:5000/cost-history?since=1715900000&until=1716000000"

# Benchmark with cost comparison
curl -X POST http://server:5000/test \
  -H "Content-Type: application/json" \
  -d '{"model": "all"}'
```

# 6 Use Cases for Cost Estimation

## 6.1 Pre-Migration Planning

Before committing to a cloud AI provider, run sheLLaMa for 1–4 weeks with your team’s actual workload. The costs page will show:

- Total monthly projected spend per provider

- Which provider offers the best price for your workload mix
- Whether cheaper models (Nova Micro, GPT-4o mini) are viable alternatives
- The ratio of input to output tokens in your usage (affects provider choice)

## 6.2 Budget Forecasting

The time-range filtering enables trend analysis:

- Daily token consumption patterns (peak hours, quiet periods)
- Week-over-week growth as AI adoption increases
- Per-user or per-team breakdown via API key tracking
- Seasonal patterns in development cycles

## 6.3 Provider Comparison

The 28-model comparison table answers questions like:

- “Is Bedrock cheaper than OpenRouter for our workload?”
- “How much would we save switching from Claude to Nova Pro?”
- “What’s the cost difference between GPT-4o and GPT-4o mini for our use case?”
- “Is the 5x price premium of Claude Opus justified over Claude Sonnet?”

## 6.4 ROI Calculation for Local Infrastructure

Compare the projected cloud spend against the cost of local hardware:

| Scenario                       | Local Cost               | Cloud Cost (monthly) |
|--------------------------------|--------------------------|----------------------|
| Light usage (10K tokens/day)   | Server: \$2,000 one-time | \$0.50–\$15/mo       |
| Medium usage (100K tokens/day) | Server: \$4,000 one-time | \$5–\$150/mo         |
| Heavy usage (1M tokens/day)    | Server: \$8,000 one-time | \$50–\$1,500/mo      |
| Team (10 devs, 5M tokens/day)  | 2 servers: \$10,000      | \$250–\$7,500/mo     |

Table 4: Break-even analysis: local hardware vs. cloud AI spend

At heavy usage, local infrastructure pays for itself in 1–6 months depending on the cloud provider chosen.

## 6.5 Hybrid Strategy Optimization

sheLLaMa’s cloud fallback feature enables a hybrid approach:

1. Run most requests locally (zero marginal cost)
2. Fall back to cloud only when local quality is insufficient
3. Track actual fallback spend separately from hypothetical costs
4. Set per-key budgets to cap cloud spend: `"budget": {"max_daily": 5.00}`

## 5. Receive webhook alerts at 80% budget consumption

The costs page separates hypothetical costs (“what if everything ran on cloud”) from actual fallback spend (“what we actually paid”), giving a clear picture of savings.

## 7 Implementation Details

### 7.1 Persistent Statistics

Cost data survives service restarts:

- Cumulative token totals saved to `persisted_totals.json` every 60 seconds
- Per-IP history maintained in memory with configurable retention
- Background thread handles periodic persistence
- `/reset-cloud-costs` endpoint for admin to clear data (requires admin role)

### 7.2 Benchmark Exclusion

Tokens from `/test` benchmark requests are excluded from cost projections. This ensures the running tab reflects real usage only—not artificial load from model comparison tests.

### 7.3 Cache Awareness

When prompt caching is enabled (default 5-minute TTL), cached responses are tracked separately. The costs page shows how many tokens were served from cache—representing additional savings that would not exist on most cloud providers (which charge per request regardless of caching).

### 7.4 Live Pricing Refresh

Listing 4: Live pricing fetch from OpenRouter

```
def fetch_cloud_pricing():
    resp = requests.get('https://openrouter.ai/api/v1/models', timeout=10)
    data = resp.json().get('data', [])
    by_id = {}
    for m in data:
        p = m.get('pricing', {})
        prompt = float(p.get('prompt', '0'))
        comp = float(p.get('completion', '0'))
        if prompt > 0 or comp > 0:
            by_id[m['id']] = {'input': prompt * 1e6, 'output': comp * 1e6}
    # Map display names to live prices...
```

Bedrock pricing is fetched via the AWS CLI:

Listing 5: AWS Pricing API call

```
aws pricing get-products --service-code AmazonBedrock \
  --region us-east-1 \
  --filters Type=TERM_MATCH,Field=regionCode,Value=us-east-1
```



## 7.5 Per-Key Budget Enforcement

API keys can have spending limits that are enforced on actual cloud fallback spend:

Listing 6: API key with budget limit

```
{
  "api_keys": {
    "sk-team-dev": {
      "name": "dev-team",
      "role": "user",
      "budget": {"max_daily": 10.00},
      "rate_limit": {"rpm": 30, "tpd": 500000}
    }
  }
}
```

When the budget is 80% consumed, a webhook fires. At 100%, cloud fallback requests are rejected with HTTP 429.

## 8 Deployment for Cost Estimation

### 8.1 Minimal Setup (Single Server)

Listing 7: Quick start for cost estimation

```
# Install and start
ollama pull qwen2.5-coder:7b
pip install flask ollama pyyaml requests psutil
python backend/app.py

# Point your team's CLI clients at the server
export SHELLAMA_API=http://server:5000
export SHELLAMA_MODEL=qwen2.5-coder:7b
source cli/shellama.bash

# Use AI normally for 1-4 weeks, then check costs
curl http://server:5000/cloud-costs | python -m json.tool
```

### 8.2 Distributed Setup (Production)

For larger teams, deploy the frontend load balancer with multiple backends:

Listing 8: backends.json for multi-server deployment

```
{
  "backends": [
    {"url": "http://192.168.1.230:5000", "weight": 10, "max_model":
      "qwen2.5-coder:7b"},
    {"url": "http://192.168.1.233:5000", "weight": 10, "max_model":
      "qwen2.5-coder:14b"}
  ]
}
```

All token tracking is centralized on the frontend, regardless of which backend processes the request.

## 9 Interpreting Results

### 9.1 Reading the Cost Table

A typical `/cloud-costs` response after one week of team usage:

```
{
  "prompt_tokens": 847293,
  "response_tokens": 2341567,
  "total_tokens": 3188860,
  "pricing_source": "openrouter",
  "cloud_costs": [
    {"provider": "Claude 4 Sonnet", "total_cost": 37.66},
    {"provider": "GPT-4o", "total_cost": 25.53},
    {"provider": "Gemini 2.5 Flash", "total_cost": 6.11},
    {"provider": "Amazon Nova Micro", "total_cost": 0.36},
    {"provider": "Bedrock Claude Opus 4", "total_cost": 188.22}
  ]
}
```

**Interpretation:** This team's weekly AI usage would cost between \$0.36 (Nova Micro) and \$188.22 (Claude Opus 4) on cloud providers. The local infrastructure cost is \$0 marginal after hardware purchase.

### 9.2 Token Ratio Analysis

The ratio of prompt to response tokens affects provider choice:

- **High output ratio** (code generation, explanations): Providers with low output pricing win (Nova Micro, Llama, GPT-4o mini)
- **High input ratio** (analysis of large files, context-heavy chat): Providers with low input pricing win (Gemini with large context, Nova Lite)
- **Balanced:** Overall total cost is the deciding factor

### 9.3 Quality vs. Cost Tradeoff

The `,test` benchmark lets you compare local model quality against cloud alternatives. If a \$0.04/request local model produces acceptable output for 95% of tasks, the ROI of local infrastructure is clear. The 5% that needs cloud fallback can be budgeted precisely using the actual fallback tracking.

## 10 Limitations

- **Token counts are model-dependent:** Different tokenizers produce different counts for the same text. Local Ollama token counts are approximate proxies for cloud provider counts.
- **Pricing changes:** Live pricing is fetched per-request for benchmarks but cached otherwise. Prices may change between fetches.
- **No latency costing:** Cloud providers may offer faster responses; the cost of developer wait time is not captured.

- **No quality weighting:** A cheap model that requires 3 retries may cost more than an expensive model that succeeds first try. This is partially addressed by the fallback tracking.
- **Region-specific:** Bedrock pricing is for us-east-1. Other regions may differ.

## 11 Conclusion

sheLLaMa transforms local LLM infrastructure into a cloud AI cost estimation platform. By running actual workloads locally and projecting costs across 28+ cloud models with live pricing, organizations can make data-driven decisions about:

- Whether to adopt cloud AI services
- Which provider offers the best value for their specific workload
- When local infrastructure ROI exceeds cloud spend
- How to budget for hybrid local+cloud deployments
- Per-team and per-project cost allocation

The platform requires no cloud accounts, no API keys, and no ongoing costs to operate as a cost estimator. It runs fully offline after the initial model pull, making it suitable for air-gapped environments and organizations evaluating AI adoption without committing to vendor contracts.

## 12 References

- OpenRouter pricing API: <https://openrouter.ai/api/v1/models>
- AWS Bedrock pricing: <https://aws.amazon.com/bedrock/pricing/>
- Ollama: <https://ollama.com>
- sheLLaMa source: <https://github.com/rory/shellama>